University of Piraeus
School of Information and Telecommunications Technologies
Department of Digital Systems

Level: Postgraduate Program of Study

Network Security

Case 5

Supervisor: Prof. Christos Xenakis

Kalaitzidis Ioannis ioannis_kalaitzidis@ssl-unipi.gr

Piraeus

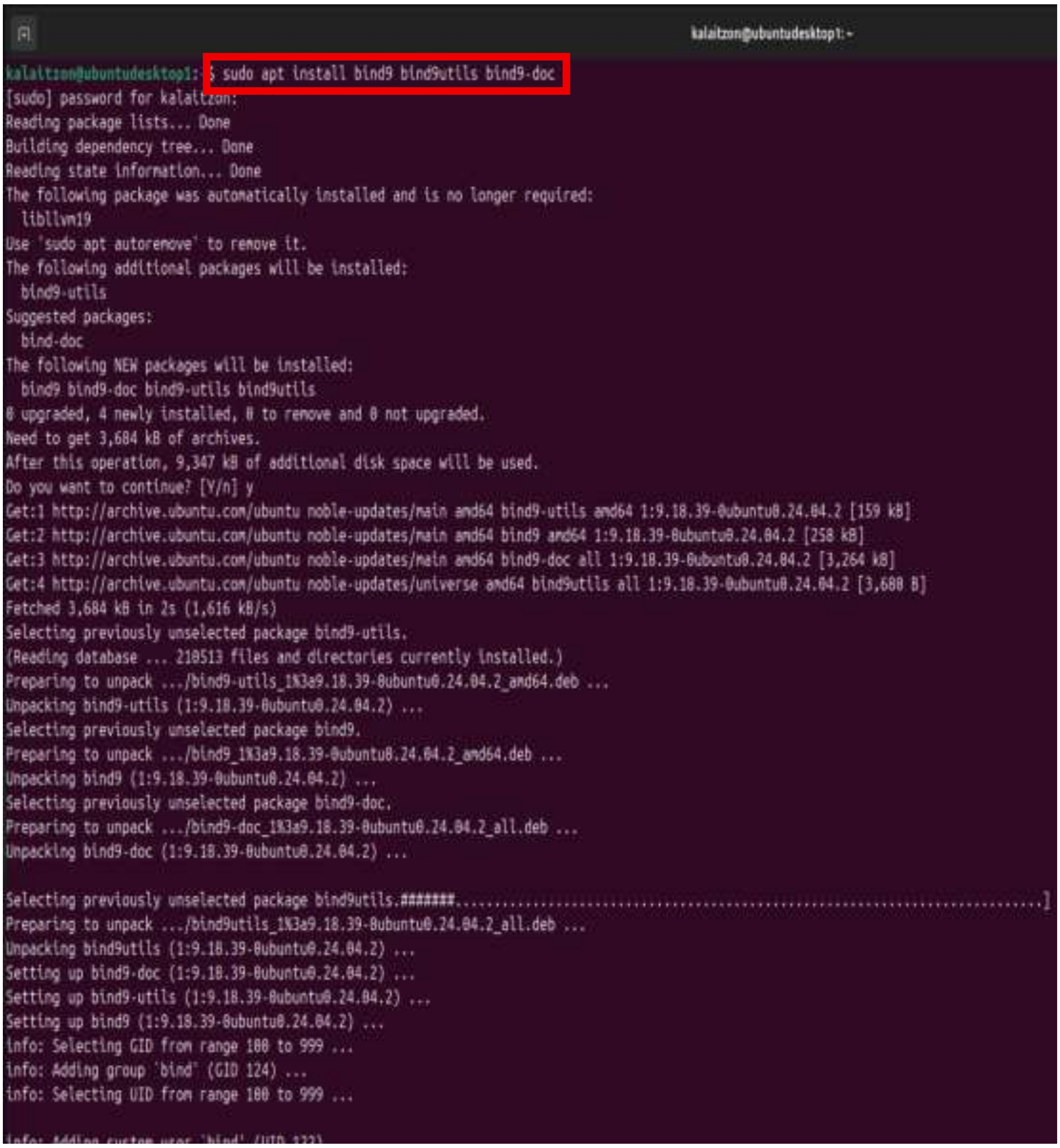
22/11/2025

Abstract

Install DNS server using the BIND9 tool on Ubuntu and create a Master Zone for the domain "Kalaitzon.gr". Set up the Forwarding to Google function and connect the domain to the server's IP. The correct operation was confirmed through a check by Windows 10 (client).

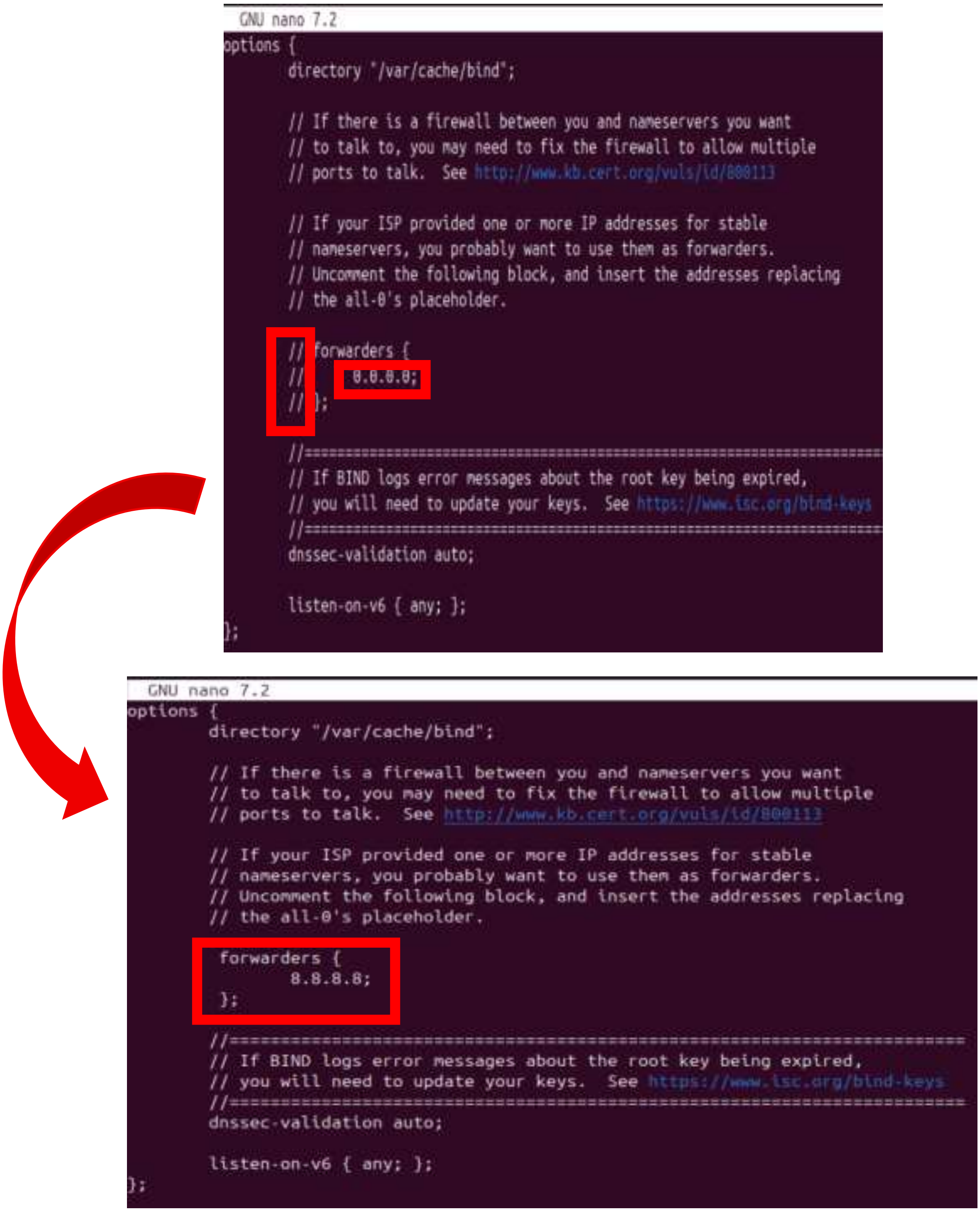
Contents

Practical piece of work

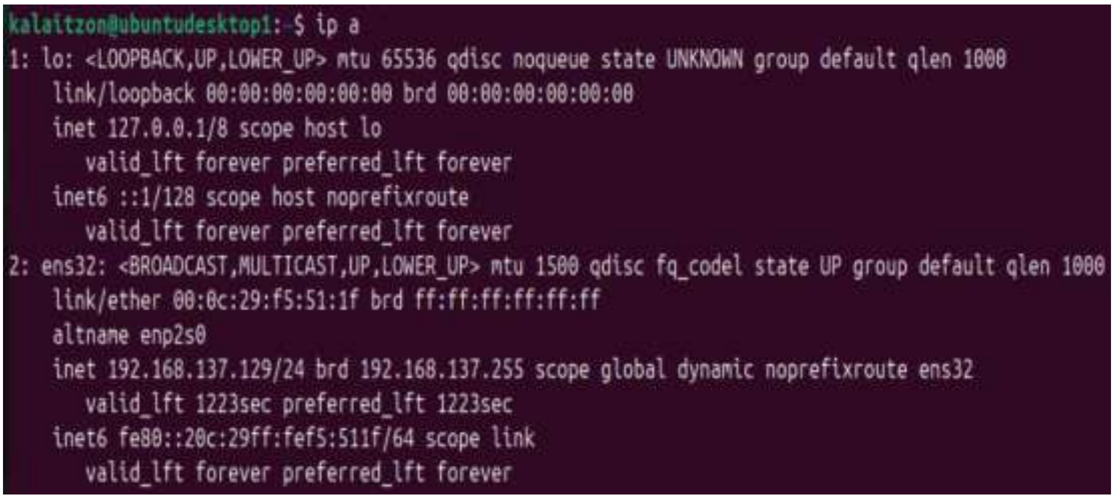


We "download" bind9 with the command **sudo apt install bind9 bind9utils bind9-doc**.

Then we set the Forwarding (Recursive Query). Without it, our server can only answer queries for our own domain (e.g. Kalaitzon.gr). With the command **sudo nano /etc/bind/named.conf.options** we open the settings file and remove the comment (//) by setting the Google IP (8.8.8.8 which is normally 008.008.008.008). We save it and exit by pressing Ctrl + O and Enter. Then Ctrl + X to return to the terminal. We do this because when the user, e.g. from Windows, asks the server for example what the IP of YouTube is, it will transfer the question to the IP of Google and in this way the server will give the answer. Under other circumstances our server would be isolated and would not know anything other than our own domain.

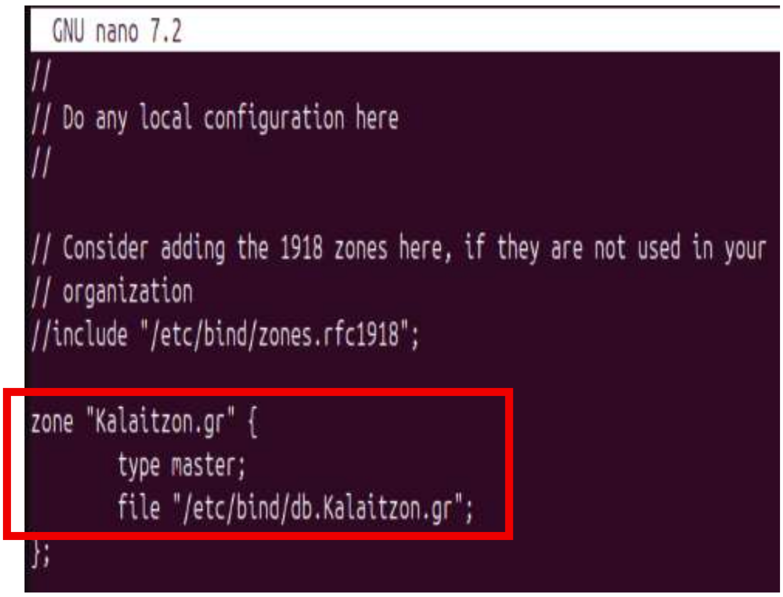


Then we will find the IP of the host with the command **ip a**. The IP of Ubuntu is **192.168.137.129**



```
kalaitzon@ubuntudesktop1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f5:51:1f brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.137.129/24 brd 192.168.137.255 scope global dynamic noprefixroute ens32
       valid_lft 1223sec preferred_lft 1223sec
    inet6 fe80::20c:29ff:fef5:511f/64 scope link
       valid_lft forever preferred_lft forever
```

Next, we proceed to create our own Local Zone. With the command **sudo nano /etc/bind/named.conf.local** we edit the configuration file to declare the domain "Kalaitzon.gr". Within the file, we define the zone type as master, thus declaring that our server will be the Authoritative Name Server for the specific domain. The creation of the Local Zone is necessary because the domain "Kalaitzon.gr" It is private and does not exist on the internet. Therefore, DNS servers, such as Google's, do not have records for this and are unable to resolve it. Through the local zone, we define our own server as the Authoritative Source for that domain. Thus, we achieve the resolution of names locally, without depending on external providers, essentially creating an internal network. At the same time, with the file command, we connect the zone to the db.Kalaitzon.gr file (which we will create below). This will be the data file that will include all the necessary information of our domain. In other words, we separate the resolution process. If a client requests an address outside of our domain, the Forwarding we previously set up is activated. On the



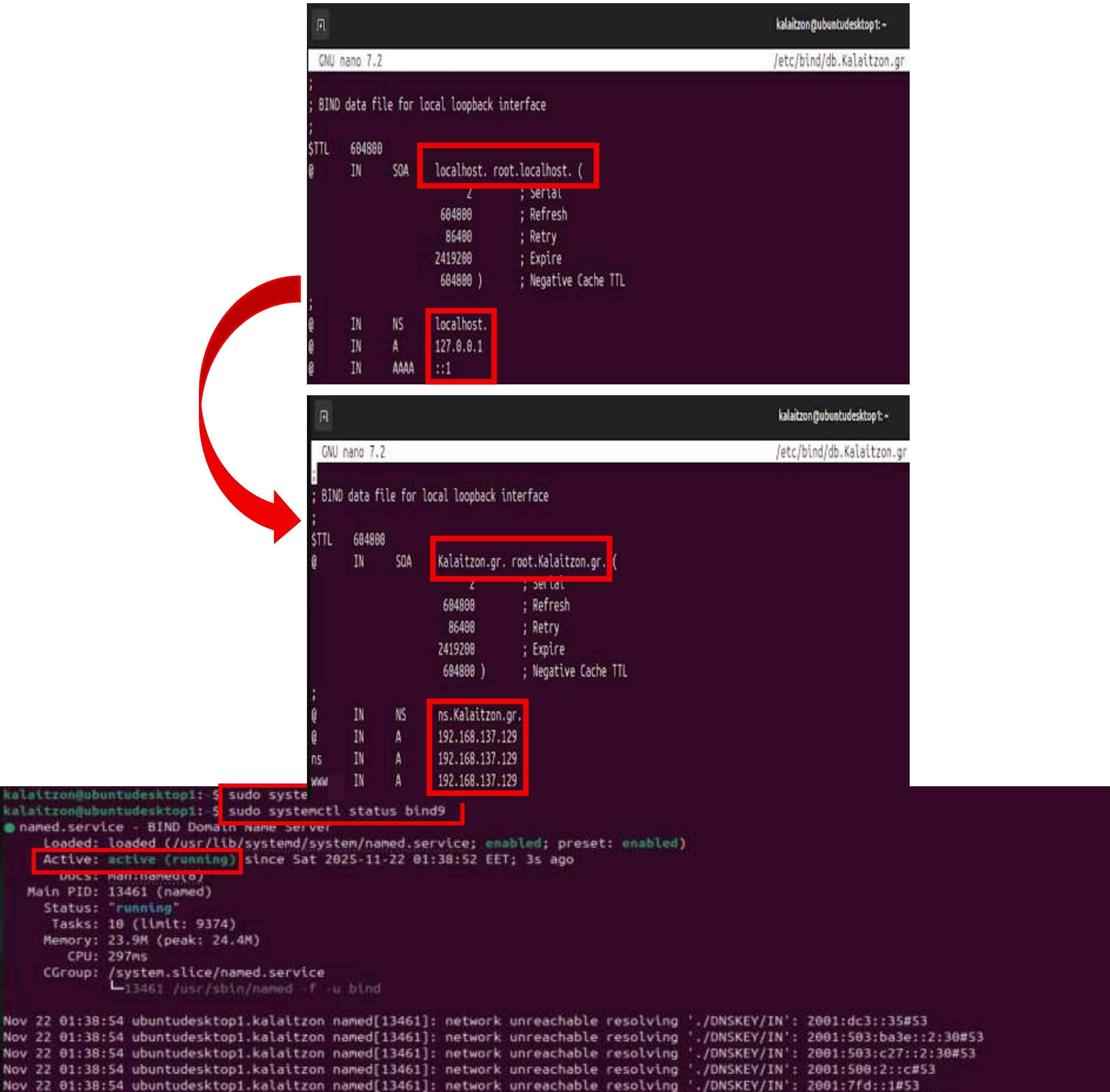
```
GNU nano 7.2
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "Kalaitzon.gr" {
        type master;
        file "/etc/bind/db.Kalaitzon.gr";
};
```

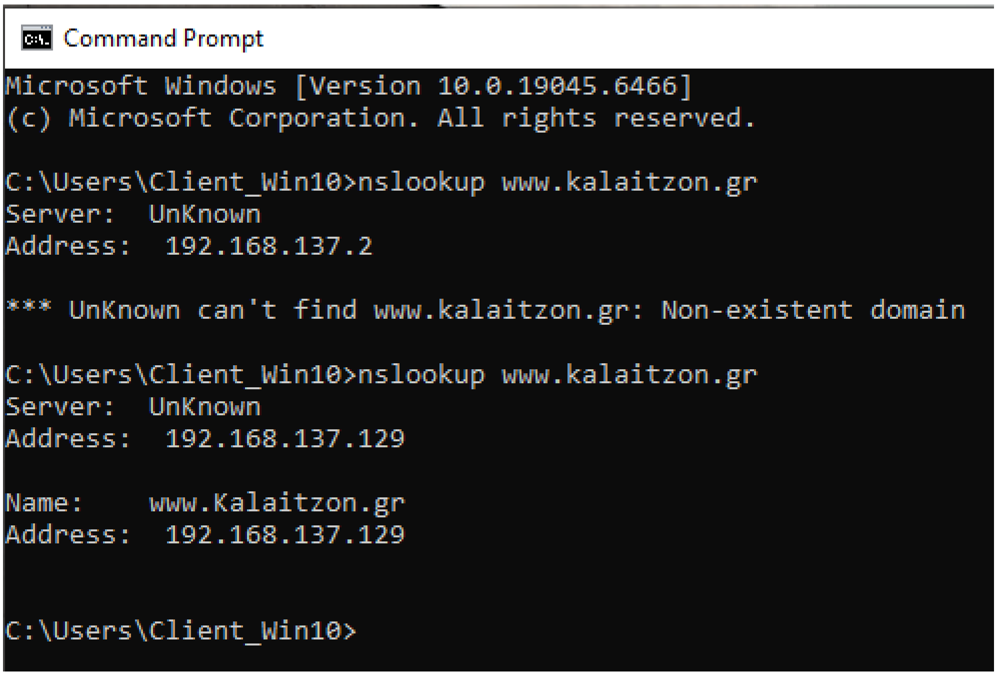
contrary, for queries related to "Kalaitzon.gr", the server does not address the Google IP, but draws the information directly from the specific local file.

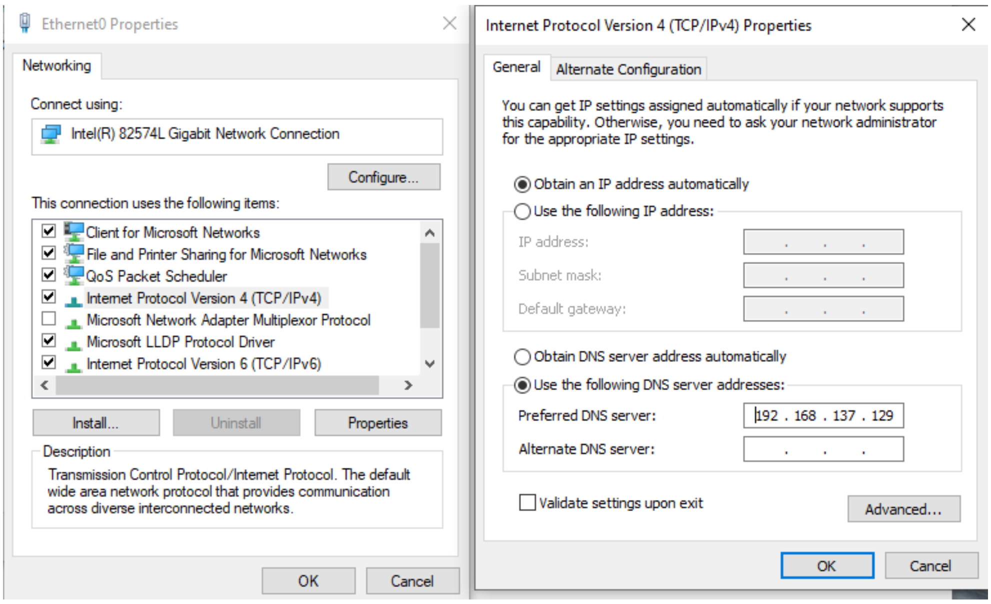
Now we will create the file that we declared in the zone above. With the command **sudo cp /etc/bind/db.local /etc/bind/db. Kalaitzon.gr** we will copy the template file that exists from localhost to ours (db.Kalaitzon.gr) and then edit it with the **sudo nano /etc/bind/db command. Kalaitzon.gr**. Where it has localhost, we replace it with Kalaitzon.gr (our domain) and where we see IP we put the one of Ubuntu (**192.168.137.129**).



We do the standard typing to save the file and proceed to restart the server with the command **sudo systemctl restart bind9**. If it does not get an error message, then it means that it worked. To check that everything went well and it works, we will use the command **sudo systemctl status bind9** and we will observe the line that says **Active: active (running)**

To confirm the proper functioning of the DNS Server, a second virtual machine with Windows 10 operating system was used, which acted as a Client. Initially, an attempt was made to resolve the domain name www.kalaitzon.gr through the nslookup command. This attempt failed (displaying a "Non-existent domain" message) because the client was configured to use the network's default DNS server (VMware Gateway, 192.168.137.2), who did not have registrations for our local zone. To address this, the client's Network Adapter Settings have been modified. Specifically, in the TCP/IPv4 protocol, the IP address of our Ubuntu Server (192.168.137.129) was statically set as Preferred DNS Server. After applying the settings, the **nslookup command www.kalaitzon.gr** executed again. This time, the client contacted our DNS Server directly, which responded successfully, resolving the domain name to the correct IP address (192.168.137.129), thus confirming that the Authoritative Server has been configured correctly.





Bibliography

ITGLOBAL.COM NL. (2025, February 28). How to configure BIND9 DNS Server on Ubuntu 20.04 | Serverspace. https://serverspace.io/support/help/configure-bind9-dns-server-on-ubuntu/

Ps, K. (2025, May 13). BIND9 check zone file - How we do it. Bobcares. https://bobcares.com/blog/bind9-check-zone-file/

What is DNS? | How DNS works | Cloudflare. (n.d.). https://www.cloudflare.com/learning/dns/what-is-dns/

Blackburn in Tech. (2024, November 7). Setting up your own DNS server: Building a BIND9 server on Ubuntu! [Video]. YouTube. https://www.youtube.com/watch?v=6mfEYAkDwRs

Koroma Tech. (2025, July 14). How to Set Up a Local DNS Server with BIND9 | Ultimate Homelab Networking Tutorial [Video]. YouTube. https://www.youtube.com/watch?v=HJHOkZb1bQQ

Artemis. (2013, November 5). How do I do a complete BIND9 DNS Server Configuration with a hostname? Ask Ubuntu. https://askubuntu.com/questions/330148/how-do-i-do-a-complete-bind9-dns-server-configuration-with-a-hostname

Bajramović, M. (2023, August 15). Configuring BIND DNS server for internal network. Medium. Retrieved from https://medium.com/@crkomarsal/configuring-bind-dns-server-for-internal-network-f853c949ddd0